Market Risk During Financial Stress

A Comparative Analysis *of Volatility, Drawdowns, Recovery and Market Sensitivity*

Independent Quantitative Finance Project

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Data period: 1 January 2020 – 31 December 2021

Abstract

This project looks at how market risk shifted during the February–March 2020 stress period, and what happened afterwards, across five series: Reliance Industries, HDFC Bank, Infosys, Tata Motors and the NIFTY 50 benchmark. Using daily closing prices from 1 January 2020 to 31 December 2021, I calculated annualized volatility, maximum drawdown, recovery time, return correlation, and beta. Volatility rose sharply for every series during the stress window, but the drawdowns and recovery times told a more mixed story: some stocks fell further and took much longer to recover than others, which is really the main purpose of this project as volatility alone doesn't tell you much about risk.

Research Question

How did market risk vary among the four Indian listed companies from different sectors during an important period of financial stress, and how did their performance compare to the overall NIFTY 50 market index?

1. Objectives

- Measure changes in daily-return volatility between a pre-stress period, the acute stress window, and the subsequent recovery period.
- Compare maximum drawdowns and the time required for each series to recover to its previous closing-price peak.
- Examine return correlations among the four companies to understand their co-movement.
- Measure each stock's beta and return correlation relative to the NIFTY 50 benchmark.
- Use trading volume as a supporting indicator of changes in market activity during the stress window.

2. Companies, Benchmark and Data

I picked four companies from different sectors: Reliance Industries, HDFC Bank, Infosys and Tata Motors, plus the NIFTY 50 as a market-wide benchmark. The company data came from NSE historical equity extracts; a couple of the files had more than one series listed, so I kept only the normal EQ series and dropped the rest. The Tata Motors data is the NSE series downloaded under the symbol TMPV. It covers the company as it traded in 2020–2021, before the 2025 demerger. The NIFTY 50 file has daily Date, Open, High, Low, Close, Shares Traded and Turnover columns and covers the full 2020–2021 window I needed.

Table 1: Companies, benchmark and data used

Series	Role in study	Data used
Reliance Industries (RELIANCE)	Company	NSE EQ daily data
HDFC Bank (HDFCBANK)	Company	NSE EQ daily data
Infosys (INFY)	Company	NSE EQ daily data
Tata Motors (TMPV)	Company	NSE EQ daily data
NIFTY 50	Market benchmark	Daily index OHLC and activity data

3. Period Definition

The same dates are used for all five series:

Table 2: Study periods and their purpose

Period	Dates	Purpose
Pre-stress	1 Jan 2020 – 19 Feb 2020	Baseline market behaviour
Stress	20 Feb 2020 – 23 Mar 2020	Acute COVID-19 market sell-off
Recovery	24 Mar 2020 – 31 Dec 2021	Post-trough period

4. Methodology

Daily return: Daily returns were calculated as the percentage change between consecutive closing prices:

Daily Return = (Current Closing Price − Previous Closing Price) / Previous Closing Price.

Annualized volatility: Sample standard deviation of daily returns × √252.

Maximum drawdown: Largest percentage decline from a historical closing-price peak to a later trough during 2020–2021.

Recovery time: Calendar days from the maximum-drawdown trough to the first later date when the closing price returned to or exceeded the previous peak.

Correlation: Pearson correlation of daily returns on common trading dates.

Beta: Covariance of a stock's daily returns with NIFTY 50 daily returns divided by the variance of NIFTY 50 daily returns, calculated using common trading dates.

5. Volatility Results

Annualized volatility of the NIFTY 50 spiked from 14.5% before the stress period to 66.0% during the stress period - a major spike for any index. This happened consistently for all four stocks. However, Tata Motors is already

volatile even before the crisis; its volatility before the stress period is 52.8% driven by large daily moves in early January, compared to Reliance, HDFC Bank and Infosys' volatility of 18.4% to 24.4%.

Table 3: Annualized volatility by period

Asset	Pre-stress	Stress	Recovery
NIFTY 50	14.5%	66.0%	20.4%
Reliance	24.4%	85.4%	34.1%
HDFC Bank	18.4%	65.1%	31.6%
Infosys	21.0%	66.4%	29.3%
Tata Motors	52.8%	82.3%	52.9%

6. Drawdown and Recovery

The highest drawdowns for 2020–2021 were 38.4% for the NIFTY 50, 44.1% for Reliance, 40.5% for HDFC Bank, 34.2% for Infosys, and 67.4% for Tata Motors, about 1.75 times the index's decline. The recovery periods are even more dispersed. In contrast to HDFC Bank (227 days), the NIFTY 50 (231 days), and particularly Tata Motors (283 days), Reliance recovered to its previous top in just 74 days. Infosys arrived at 114 days in between.

Table 4: Maximum drawdown, peak, trough and recovery time

Asset	Max drawdown	Peak	Trough	Recovery days
NIFTY 50	38.4%	14-Jan-2020	23-Mar-2020	231
Reliance	44.1%	17-Jan-2020	23-Mar-2020	74
HDFC Bank	40.5%	14-Jan-2020	24-Mar-2020	227
Infosys	34.2%	19-Feb-2020	23-Mar-2020	114
Tata Motors	67.4%	15-Jan-2020	03-Apr-2020	283

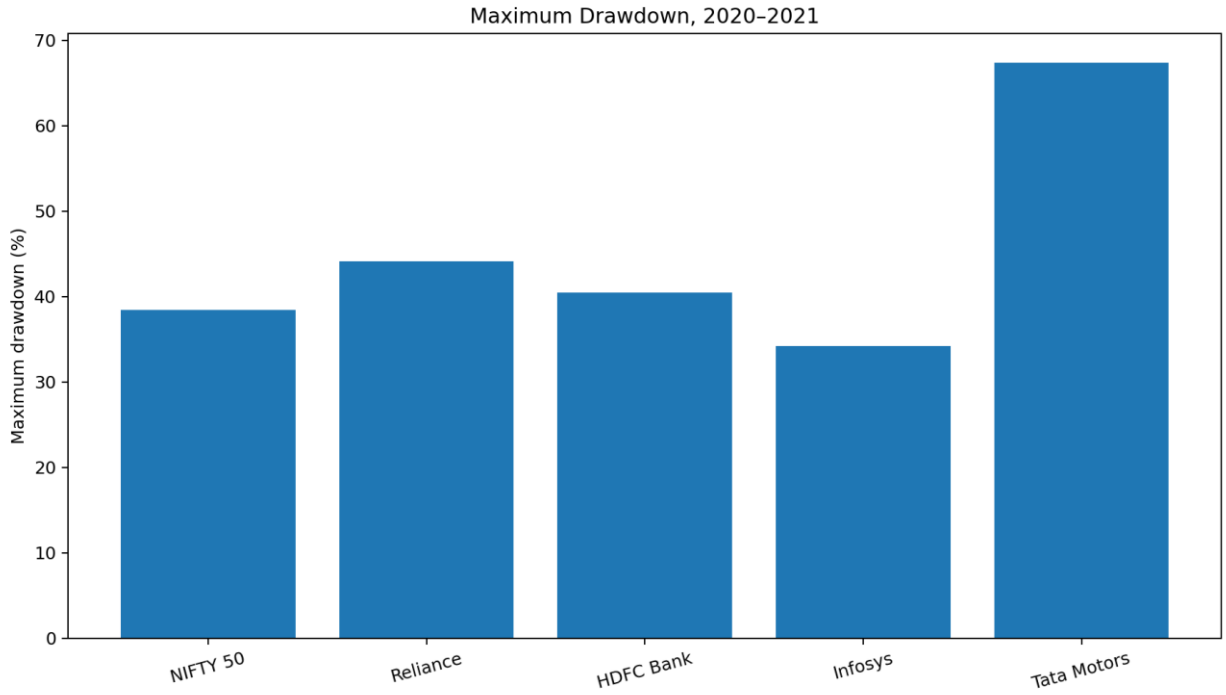


Figure 1: Maximum drawdown, 2020-2021

7. Market Sensitivity to NIFTY 50

Bringing in the NIFTY 50 lets me check how each stock moved relative to the wider market. With betas of 1.096 for Reliance, 1.094 for HDFC Bank, 0.835 for Infosys, and 1.386 for Tata Motors for the entire 2020–2021 sample, three of the four stocks moved more strongly than the index, with Infosys being the only one that was somewhat less sensitive. The correlations add something the betas don't show: Reliance (0.717) and HDFC Bank (0.805) followed the market closely, but Tata Motors, with the highest beta (1.386), had a correlation of only 0.612. Its swings were bigger but less explained by the overall market.

Another thing I noticed is that all four stocks moved much more in step with the NIFTY 50 during the crash. Their correlations rose to 0.83-0.90 in the stress period, up from 0.61-0.80 over the full period. This suggests that diversification may protect less in a crisis, although with only 21 trading days in the stress window, I treat this as an indication rather than a firm conclusion. The full-period figure includes the stress days, so this comparison is indicative only.

Table 5: Beta and correlation with the NIFTY 50 (full period and stress period)

Asset	Beta	Correlation	Stress beta	Stress correlation
Reliance	1.096	0.717	1.168	0.903
HDFC Bank	1.094	0.805	0.814	0.827
Infosys	0.835	0.648	0.901	0.896
Tata Motors	1.386	0.612	1.064	0.854

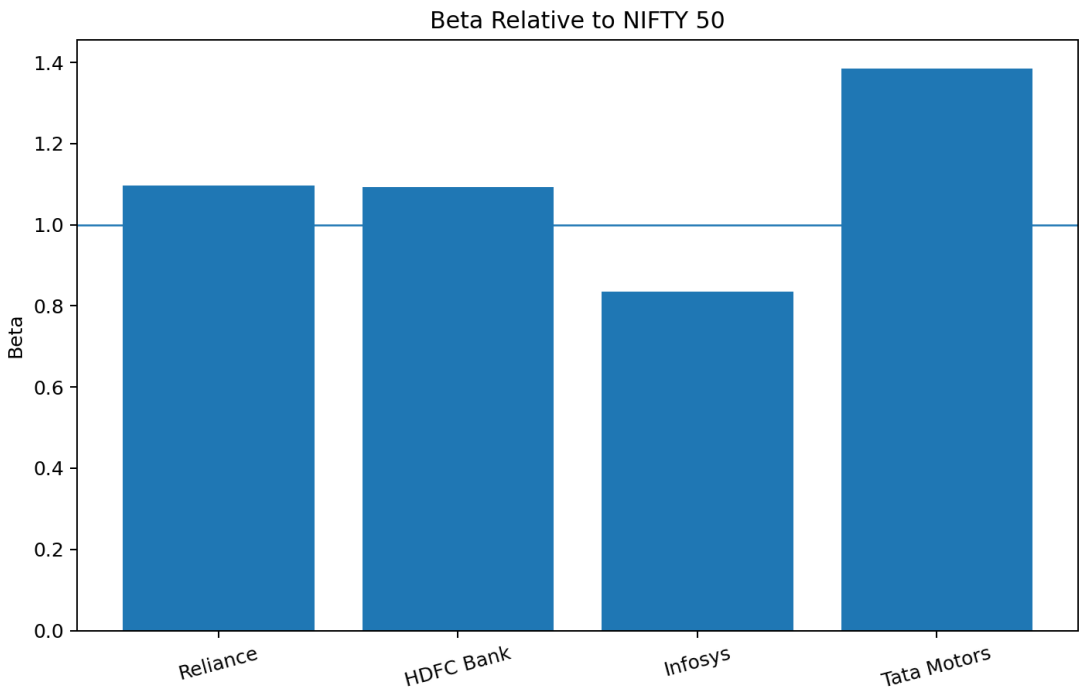


Figure 2: Beta relative to the NIFTY 50

8. Price Path and Market Activity

The chart below normalizes every series to 100 at its first close on 1 January 2020, so the price paths are comparable even though the index and share prices are on completely different scales. The shaded band marks the acute stress window.

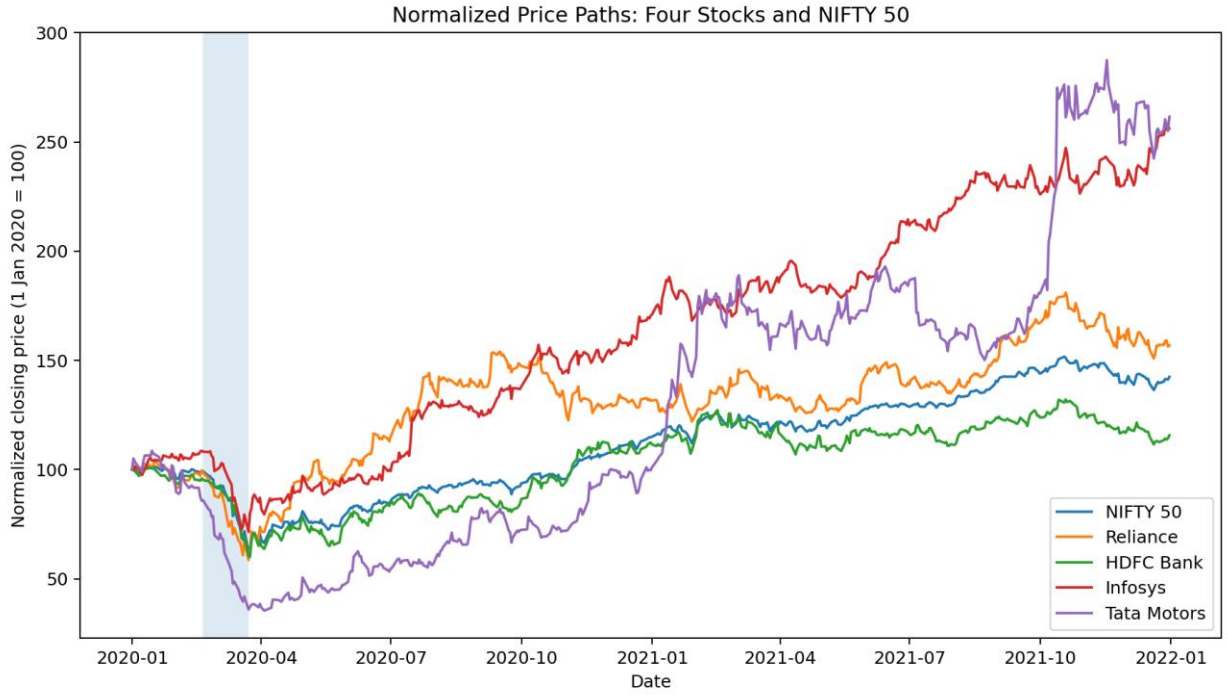


Figure 3: Normalized price paths of the four stocks and the NIFTY 50 (1 Jan 2020 = 100)

Trading volume picked up for all four stocks during the stress window compared with the pre-stress period, which is not surprising given how much prices were moving. The NIFTY 50 file does include its own shares-traded and turnover numbers, but I kept those separate rather than comparing index activity directly against individual-stock volumes, since the two aren't really measuring the same thing.

Table 6: Average daily trading volume, pre-stress vs stress period

Stock	Pre-stress avg. daily volume	Stress avg. daily volume	Change
Reliance	8.39 million	19.03 million	+127%
HDFC Bank	5.92 million	17.89 million	+202%
Infosys	6.92 million	12.14 million	+76%
Tata Motors	43.30 million	78.77 million	+82%

9. Return Correlation

Among the four companies, Reliance and HDFC Bank had the highest pairwise correlation at 0.496, and Infosys and Tata Motors the lowest at 0.352. All the correlations are positive but well below 1, meaning these stocks moved somewhat together but far from identically. Over the full period, that’s a useful diversification signal, although correlations were higher during the stress window (see Section 7). Obviously, correlation alone isn’t enough to build a portfolio recommendation.

Table 7: Pairwise correlation of daily returns among the four stocks

Asset	Reliance	HDFC Bank	Infosys	Tata Motors
Reliance	1.000	0.496	0.415	0.394
HDFC Bank	0.496	1.000	0.411	0.432
Infosys	0.415	0.411	1.000	0.352
Tata Motors	0.394	0.432	0.352	1.000

10. Key Findings

- Volatility increased sharply in the stress period for the NIFTY 50 and all four stocks.
- Stock drawdowns ranged from 34.2% (Infosys) to 67.4% (Tata Motors). Tata Motors and Reliance fell more than the NIFTY 50 (38.4%), HDFC Bank fell by a similar amount, and Infosys fell less.
- Recovery took anywhere from 74 days (Reliance) to 283 days (Tata Motors) - a huge spread.
- Beta came in above 1 for Reliance, HDFC Bank and Tata Motors, and below 1 only for Infosys.
- Pairwise correlations among the four stocks stayed positive but modest, from 0.352 to 0.496: real co-movement, but nowhere close to 1.

- Correlations with the NIFTY 50 rose sharply during the stress period (0.83 to 0.90), suggesting that diversification protects less in a crisis.

11. Interpretation for Risk Management

The point I want to make here is that there is no one statistic that can give you the entire risk picture. Volatility measures the dispersion of daily return, drawdown measures the magnitude of loss from a prior high, and recovery time measures the duration for which you would be holding on to this loss. Beta measures the sensitivity to the market. Ignoring any of these would be ignoring what the other measure(s) capture.

NIFTY 50 volatility jumped during the crash, but the four stocks didn't respond the same way. They differed in beta and in how far they fell.

12. Limitations

- Four companies and one stress episode is a small sample which isn't meant to represent the whole Indian equity market.
- I'm working from end-of-day closing prices, so anything that happened intraday is invisible here.
- Beta shifts depending on the sample period and benchmark chosen, so these estimates aren't fixed values.
- As always with historical data, past returns and risk measures don't guarantee anything about future performance.
- Raw closing prices can be distorted by corporate actions like splits or bonuses. A more rigorous version of this study would use adjusted prices instead.
- Since each company is just one firm in its sector, none of these results should be generalized into sector-wide conclusions.
- The stress period has only 21 trading days and the pre-stress period only 36, so volatility and stress-period beta estimates are noisy. The stress window was also chosen after the fact, with knowledge of the crash dates.
- The Tata Motors series is the pre-demerger company, including Jaguar Land Rover, so it doesn't represent today's commercial vehicles business.

13. Conclusion

All the five series showed an increase in their level of volatility in the February-March 2020 market crash. However, there was more to come as each stock series suffered from unique levels of drawdown duration and magnitude. The inclusion of the NIFTY 50 in the portfolio made it possible for me to gauge this aspect of sensitivity using beta and correlation. In summary, a consideration of volatility alone meant missing out on much else in these five series.

14. Data and References

Primary data: NSE historical daily equity data for Reliance Industries, HDFC Bank, Infosys and Tata Motors, covering 1 January 2020 to 31 December 2021. Data were downloaded from the NSE India website in September 2026.

Primary benchmark data: NIFTY 50 historical daily data obtained from the official NIFTY Indices historical-data website for 1 January 2020 to 31 December 2021, including Open, High, Low, Close, Shares Traded and Turnover.

Data handling: Company datasets were filtered to the EQ series, sorted by date and checked for duplicate dates before calculating returns. NIFTY 50 returns were calculated from consecutive closing values. Beta and benchmark correlations were calculated using the available common trading dates.

Data Sources: Historical daily equity data for Reliance Industries, HDFC Bank, Infosys and Tata Motors were obtained from the National Stock Exchange of India (NSE). Historical NIFTY 50 index data were obtained from the official NIFTY Indices website. The study covers 1 January 2020 to 31 December 2021.

NSE: https://www.nseindia.com/all-reports#cr_equity

NIFTY Indices: https://www.niftyindices.com/reports/historical-data